

University of Minnesota
School of Physics and Astronomy

2026 Fall Mathematics 8302
Differential Topology
Assignment Solution

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Question 1.7

Let N denote the **north pole** $(0, \dots, 0, 1) \in \mathbb{S}^n \subseteq \mathbb{R}^{n+1}$, and let S denote the **south pole** $(0, \dots, 0, -1)$. Define the **stereographic projection** $\sigma : \mathbb{S}^n \setminus \{N\} \rightarrow \mathbb{R}^n$ by

$$\sigma(x_1, \dots, x_{n+1}) = \frac{(x_1, \dots, x_n)}{1 - x_{n+1}}. \quad (1)$$

Let $\tilde{\sigma} = -\sigma(-x)$ for $x \in \mathbb{S}^n \setminus \{S\}$.

- (a) For any $x \in \mathbb{S}^n \setminus \{N\}$, show that $\sigma(x) = u$, where $(u, 0)$ is the point where the line through N and x intersects the linear subspace where $x_{n+1} = 0$. Similarly, show that $\tilde{\sigma}(x)$ is the point where the line through S and x intersects the same subspace. (For this reason, $\tilde{\sigma}$ is also called **stereographic projection from the south pole**.)
- (b) Show that σ is bijective and

$$\sigma^{-1}(u^1, \dots, u^n) = \frac{(2u^1, \dots, 2u^n, |u|^2 - 1)}{|u|^2 + 1}. \quad (2)$$

- (c) Compute the transition map $\tilde{\sigma} \circ \sigma^{-1}$ and verify that the atlas consisting of the two charts $(\mathbb{S}^n \setminus \{N\}, \sigma)$ and $(\mathbb{S}^n \setminus \{S\}, \tilde{\sigma})$ defines a smooth structure on $\mathbb{S}^n$. (The coordinates defined by σ and $\tilde{\sigma}$ are called **stereographic coordinates**.)
- (d) Show that this smooth structure is the same as the one defined in Example 1.31.

Example 1.31 We put a smooth structure on $\mathbb{S}^n$ as follows. For each $i = 1, \dots, n+1$, let $(U_i^\pm, \phi_i^\pm)$ denote the graph coordinate charts, where

$$U_i^\pm = \{(x_1, \dots, x_{n+1}) \in \mathbb{S}^n : \pm x_i > 0\}, \quad \phi_i^\pm(x_1, \dots, x_{n+1}) = (x_1, \dots, \hat{x}_i, \dots, x_{n+1}), \quad (3)$$

where the hat indicates that the i -th coordinate is omitted.

For any distinct indices i, j , the transition map $\phi_i^\pm \circ (\phi_j^\pm)^{-1}$ is easily computed. In the case $i < j$, we get

$$\phi_i^\pm \circ (\phi_j^\pm)^{-1}(u^1, \dots, u^n) = (u^1, \dots, \hat{u}^i, \dots, \pm \sqrt{1 - |u|^2}, \dots, u^n) \quad (4)$$

with the $\pm \sqrt{1 - |u|^2}$ in the j -th position, and a similar formula holds for $i > j$. In the case $i = j$, an even simpler formula holds:

$$\phi_i^\pm \circ (\phi_i^\mp)^{-1}(u^1, \dots, u^n) = \text{ID}_{\mathbb{B}^n}. \quad (5)$$

Thus, the collection of charts $\{(U_i^\pm, \phi_i^\pm)\}$ is a smooth atlas, and so defines a smooth structure on $\mathbb{S}^n$.

Answer

Question 1.9

Complex projective n -space, denoted by $\mathbb{CP}^n$, is the set of all 1-dimensional complex-linear subspaces of $\mathbb{C}^{n+1}$, with the quotient topology inherited from the natural projection $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$. Show that $\mathbb{CP}^n$ is a compact $2n$ -dimensional topological manifold, and show how to give it a smooth structure analogous to the one we constructed for $\mathbb{RP}^n$. (We use the correspondence

$$(x^1 + iy^1, \dots, x^{n+1} + iy^{n+1}) \longleftrightarrow (x^1, \dots, x^{n+1}, y^1, \dots, y^{n+1}) \quad (6)$$

to identify $\mathbb{C}^{n+1}$ with $\mathbb{R}^{2n+2}$.)

Answer